Rising Temperatures Drive Higher Prevalence of Chronic Kidney Diseases in the United States

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BACKGROUND

Chronic kidney diseases (CKDs) affect ~10% of the adult population in the world, and 1 in 7 adults in the US, causing 1.2 million deaths and 28 million years of life lost each year. Studies in developing countries attributed CKDs to heat and pesticide exposure, and concluded that CKDs are prevalent among agricultural workers in rural areas. However, such propositions have not been tested in the US.

INNOVATION

- This study adopts an interdisciplinary approach by incorporating heat exposure into the perspective of social determinants of health.
- This study identifies hot and cold spots and driving forces of CKDs prevalence at the county level in the US.

OBJECTIVE

To identify **spatial patterns** and **driving forces** of CKDs prevalence in the US.

METHODS

Data: CKDs prevalence data from 2009 to 2021 come from the Kidney Disease Surveillance System.

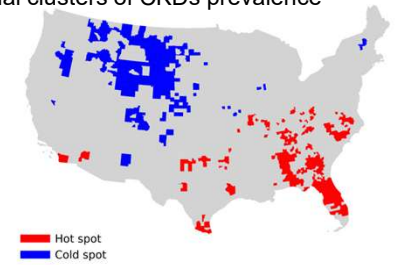
Temperature measure is sourced from the Parameter-elevation Regressions on Independent Slopes Model, a downscaled high-resolution (0.04-degree resolution, approximately 4 km^2 at the equator) climate dataset.

County-level sociodemographic characteristics come from the American Community Survey 5-year estimates from 2009 to 2021.

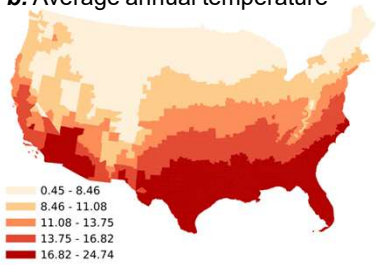
Methods: We employed spatial cluster analysis to identify hot and cold spots of CKDs prevalence and negative binomial regression to explore its driving forces.

RESULTS

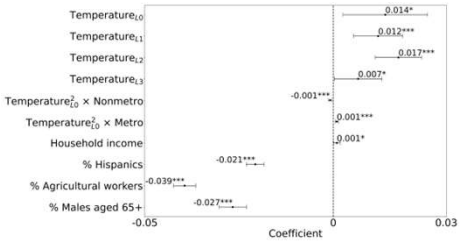
a. Spatial clusters of CKDs prevalence



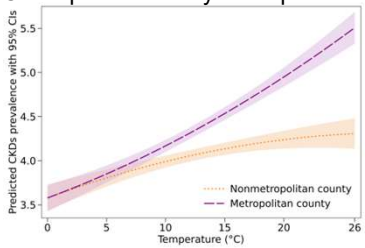
b. Average annual temperature



c. Coefficient from negative binomial regression



d. CKDs prevalence by metropolitan status



CONCLUSIONS

- Rising temperatures were associated with increased CKDs in the current year (L_0) and demonstrated long-lasting effects from preceding years (L_1, L_2, and L_3).
- Risk factors identified in developing countries (e.g., Hispanics, agricultural workers, and older males) exhibited opposite effects in the US context.
- Urban counties experienced a steeper increase in CKD prevalence in response to rising temperatures compared with rural counties.

SCAN ME

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